



Metaheuristically optimized deep soft-voting ensemble for explainable and resource-aware signal processing in wireless sensor network intrusion detection

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Abstract

Wireless Sensor Networks (WSN) are widely employed in various sensitive areas; however, due to their limited computing power, energy, and memory, these networks are exposed to signal-level anomalies as well as cyber-attacks. This study aims to introduce a metaheuristically optimized deep soft-voting ensemble that integrates Deep Neural Networks and CATBoost classifiers to enable secure intrusion detection. To achieve a seamless integration between convergence accuracy and exploration divergence, the authors have employed two complementary metaheuristics, Quadratic Interpolation Optimization (QIO) and Osprey Optimization Algorithm (OOA), for tuning the hyperparameters and decision thresholds. The system derives 23 operational and topological signal features that represent normal traffic and four major attack types, i.e., Blackhole, Grayhole, Flooding, and TDMA Scheduling. Rigorous preprocessing, such as Variance Inflation Factor analysis, supports that the features are independent and stable. The metrics used include accuracy, precision, recall, F1-score, specificity, and AUC, with the QIO-enabled model (DCQI) leading the way with the highest test accuracy of 95.62%. DCQI outperforms the baseline (DNCA) and the OOA-enhanced models, and thus, both class-level detection and balanced performance can be achieved. The feature sensitivity analysis based on the Cosine Amplitude Method and ranking both distinctly and consistently indicates Is CH, Who CH, and Dist to CH as the most influential predictors, which not only strengthens the understanding of WSNs but also shows the computational efficiency of the adopted security.

Keywords Wireless sensor networks · Intrusion detection system · Signal processing · Deep neural network · Soft voting ensemble · Feature sensitivity analysis

1 Introduction

Wireless Sensor Networks (WSNs) have emerged as one of the most transformative paradigms in modern communication and computing systems. They are widely deployed in diverse real-world applications, including environmental monitoring, industrial automation, disaster management, healthcare, and military surveillance [1]. A typical WSN consists of multiple energy-efficient, cost-effective sensor nodes that capture environmental changes, process the data locally, and transmit it wirelessly to a sink node or base station for further analysis. Depending on the deployment scenario,

sensor nodes may be equipped with thermal, acoustic, chemical, optical, or pressure sensors, enabling broad functionality across domains. Despite their versatility, WSNs face severe challenges due to inherent hardware constraints, including limited energy capacity, restricted memory, and low computational power [2, 3].

These vulnerabilities make WSNs highly susceptible to security breaches, particularly when deployed in open and potentially hostile environments. Attackers can exploit these weaknesses by compromising nodes, eavesdropping on communications, injecting false data, tampering with message integrity, or exhausting network resources [4]. Among such threats, Denial-of-Service (DoS) attacks are particularly damaging, as they disrupt network functionality in multiple ways.

Intrusion Detection Systems (IDSs) are therefore essential to safeguard WSNs. An effective IDS must not only detect known attacks but also adapt to previously unseen threats,

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issuing timely alerts upon observing abnormal behavior [5, 6]. However, designing IDSs for WSNs is difficult due to several constraints: the lack of specialized training datasets, the need for high detection accuracy, and the demand for lightweight implementations that minimize computational and energy overhead [7]. This has motivated the integration of machine learning (ML) and deep learning (DL) techniques, which are increasingly employed to enhance IDS adaptability, detection capability, and efficiency in resource-constrained WSN environments.

Several papers have been written on the implementation of ML concepts to address the localization problem in wireless sensor networks (WSNs). One of the energy-efficient localization techniques based on ML was the UWB frequency spectrum combined with angle of arrival (AOA) [8] and time difference of arrival (TDOA) [9] methods. The study by Ahmadi et al. [10] highlighted that different sensor types with specific technical features could be leveraged to enhance ML-based localization in WSNs. Although such methods provide high accuracy, their heavy dependence on specialized hardware increases deployment costs, motivating the search for more affordable alternatives.

Multiple machine learning-based methods have been proposed to address the limitations mentioned. For example, Kim et al. [11] employed Support Vector Regression (SVR) to design an RSSI-based localization algorithm that improved robustness and precision. Similarly, a three-layer feed-forward Artificial Neural Network (ANN) was reported to achieve small error rates for indoor localization [12]. The use of ANNs has been extensive. While an RSSI and KNN profiling approach was characterized as having a low cost and good estimation accuracy, a Bayesian-based plan for large-scale Wireless Sensor Networks (WSNs) was introduced as the first in [13]. Merhi et al. [14] combined decision tree (DT) classifiers with TDOA features to develop an acoustic target localization method. Furthermore, the Event-Based MAC (EB-MAC) protocol for efficient target acquisition in acoustic WSNs was also their contribution.

Optimization methods have also been widely adopted to improve localization further. For instance, Lee et al. [15] integrated particle swarm optimization (PSO) with RSSI-based models for indoor target tracking, while Wang et al. [16] leveraged probability distributions of node distances observed by anchor nodes. Cao and Xu [17] enhanced the classical DV-Hop algorithm using optimum anchor node subsets (OANS DV-Hop), employing both binary and continuous PSO variants for improved precision. In another study, TDOA and Kalman filtering were combined with DV-Hop, while ensemble optimization techniques further reduced beacon deployment errors and localization inaccuracies [18]. These approaches often achieved localization errors of less than 2%, but they remain limited by computational demands

and challenges in large-scale deployments. Table 1 shows the comparison of the recent articles with the present study.

This study builds on these directions and introduces a metaheuristically optimized deep soft-voting ensemble that combines Deep Neural Networks (DNNs) and CATBoost classifiers to strengthen intrusion detection in WSNs. The framework leverages two complementary metaheuristic optimizers, Quadratic Interpolation Optimization (QIO) and the Osprey Optimization Algorithm (OOA), to fine-tune hyperparameters and decision thresholds. A comprehensive dataset of 23 operational and topological features, collected under both normal conditions and diverse attack scenarios (Blackhole, Grayhole, Flooding, and TDMA Scheduling), is employed to validate the framework. The main contributions of this work are as follows:

- i. *Hybrid voting meta-ensemble architecture* Development of a soft-voting ensemble that integrates Deep Neural Networks (DNNs) for nonlinear feature abstraction with CATBoost classifiers for effective handling of tabular and categorical data. This combination leverages the complementary strengths of deep representation learning and gradient-boosted decision trees, resulting in expressive yet stable decision boundaries.
- ii. *Metaheuristic optimization for hyperparameter tuning* Incorporation of two complementary population-based optimization algorithms, Quadratic Interpolation Optimization (QIO) for rapid convergence and exploitation, and Osprey Optimization Algorithm (OOA) for exploratory search—ensuring robust hyperparameter tuning, avoidance of premature convergence, and improved decision thresholds.
- iii. *Feature-rich dataset design* Compilation of a dataset containing 23 network attributes, covering both normal traffic and four major attack scenarios (Blackhole, Grayhole, Flooding, and TDMA Scheduling). Multicollinearity issues are addressed using the Variance Inflation Factor (VIF) analysis, thereby enhancing model stability and interpretability.
- iv. *Comprehensive evaluation metrics* Performance assessment across accuracy, precision, recall, F1-score, specificity, and AUC, with both aggregate and per-class evaluations to ensure balanced performance across different attack types.
- v. *Explainable sensitivity and feature importance analysis* Application of Cosine Amplitude Method (CAMs) and sensitivity ranking to highlight key predictors such as *Is CH*, *Who CH*, and *Dist to CH*. This enhances explainability and strengthens trust in model decisions.
- vi. *Empirical validation of optimization efficacy* Experimental results demonstrate that the QIO-enhanced ensemble (DCQI) consistently outperforms both the baseline ensemble (DNCA) and the OOA-enhanced

Table 1 Comparison of state-of-the-art IDS approaches in WSNs and the research gap addressed by this study

References	Objective	Advantages	Limitations	Research gap	How this study fills the gap
Gupta et al. [1]	Multi-objective ML framework for WSN optimization	Balances performance trade-offs; energy efficient	Lacks deep integration of explainability; limited attack diversity	Missing explainable ensemble optimization for IDS	Introduces metaheuristically optimized soft-voting ensemble with explainability via CAM and feature ranking
Talukder et al. [2]	Hybrid ML with balancing + dimensionality reduction	Improved detection with reduced bias	Offline optimization; no real-time adaptability	Gap in adaptive and resource-aware IDS	Proposes resource-aware optimization (QIO, OOA) enabling adaptability and robustness
Alanazi et al. [3]	ML-driven routing for energy-efficient 6G-WSN	Energy-aware routing optimization	Focuses on routing, not intrusion detection	Lack of IDS focus for evolving threats	Shifts focus to intrusion resilience while maintaining energy/resource awareness
Birahim et al. [4]	PSO-based explainable ensemble IDS	Adds explainability; swarm optimization	Risk of premature convergence; limited generalizability	Gap in balancing exploration & exploitation for stable IDS	Combines QIO (exploitation) + OOA (exploration) for stability + resilience
Alipio and Bures [5]	DRL-based congestion control in WSN	Learns dynamic congestion control	Focused on routing/congestion, not IDS	IDS against adaptive attacks not addressed	Provides deep IDS framework adaptable to adversarial WSN environments
Ahmed [6]	Context-aware ML-based IDS	Adds contextual information to improve detection	Limited scalability; no ensemble integration	Gap in robust ensemble with context/explainability	Adds deep-boosted ensemble with explainable AI and scalable optimization
Priyadarshi [7]	Review of ML in IoT-WSN routing challenges	Comprehensive survey; identifies key issues	Review only, no proposed framework	Identifies lack of adaptive, resource-aware IDS	Proposes novel explainable, meta-ensemble IDS directly addressing survey-highlighted issues
This study	Metaheuristically optimized deep soft-voting ensemble IDS	High detection (multi-metric), explainable AI, resource-aware, balanced exploration/exploitation, resilience to adversarial noise	Dataset still controlled; no full real-time retraining	Expands IDS research by unifying deep + boosted ensemble, explainability, resource awareness, and optimization trade-offs	Provides novel, explainable, resource-aware, and adaptive IDS framework for WSNs

model (DCOA), achieving superior detection accuracy, stability, and generalization.

In summary, this study introduces a resource-aware and explainable intrusion detection framework for WSNs, achieving high detection accuracy while maintaining computational efficiency. The proposed system is robust against diverse attack types and tailored for the constrained environments typical of WSN deployments.

2 Data collection

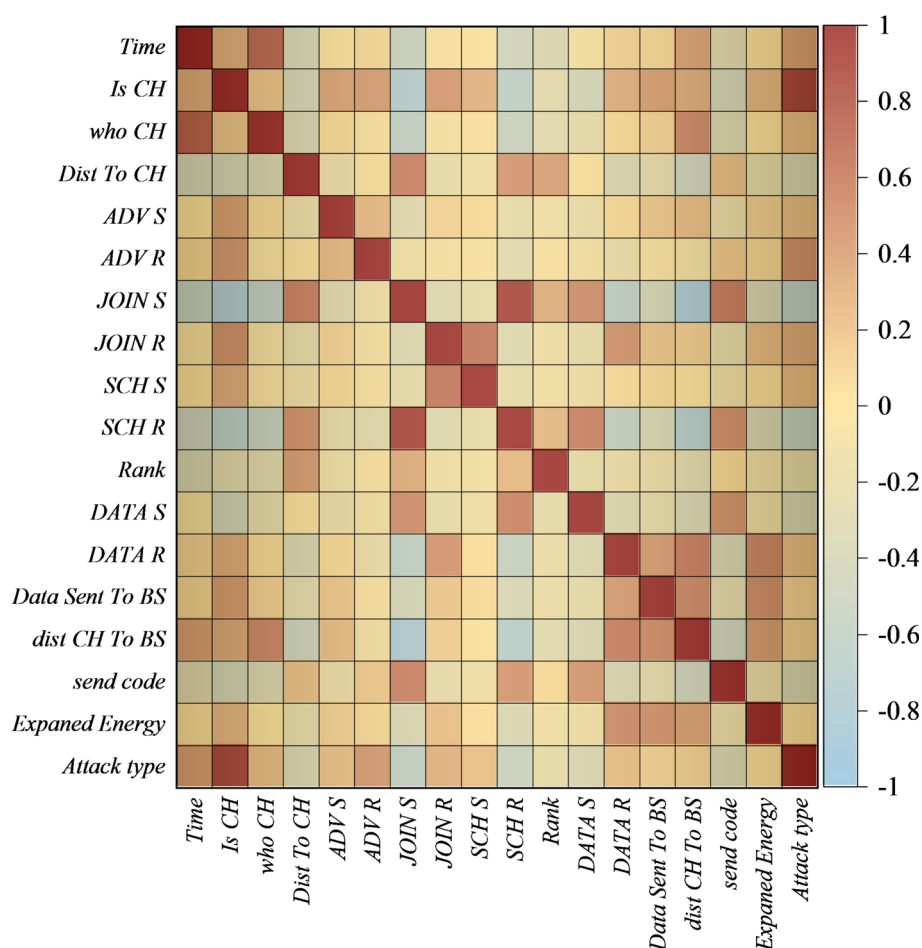
To generate the dataset and gather the necessary data from the transmitted and received packets within the WSN, a low-cost monitoring service is required. On the other hand, all relevant network data should be captured to aid in the detection, classification, and prevention of various potential assaults. In this study, to disperse the load across sensor nodes, each sensor participates in the monitoring process and must be able to monitor a collection of its neighbors. The task was determining the optimal number of nodes to observe by a sensor node in order to monitor all network sensors. The dataset was gathered from a published study, which included an experimental process for monitoring, with 374,661 samples, and 23 attributes were extracted to identify each node's status in the network [19]. The dataset is listed in Table 2.

Figure 1 illustrates the correlation matrix among the variables. Regarding the relationship between the Attack Type and the input features, the variable “Is CH” exhibits a strong positive correlation with the output, whereas “JOIN S” shows a strong negative correlation. Additionally, several other variables display varying degrees of positive and negative correlations, which had a key role in guiding the process of feature selection and prioritization for subsequent analysis.

Table 3 presents the description of features alongside their Variance Inflation Factor (VIF) scores for both the initial and reduced feature vectors. VIF analysis was conducted to detect multicollinearity among predictors, with higher VIF values indicating stronger correlation with other variables. In the initial feature vector, most variables had acceptable VIF values (typically below the critical threshold of 10), except for “JOIN S” (VIF = 37.1710) and “SCH R” (VIF = 27.8604), which showed extremely high multicollinearity. These two features were therefore removed in the reduced feature vector to improve model stability and avoid redundancy in prediction. After removal, the VIF scores of the remaining features generally decreased, indicating reduced multicollinearity. For example, “Dist To CH” decreased from 3.5336 to 2.3373, and “DATA S” dropped from 4.3942 to 2.7054. The reductions suggest that the elimination of highly collinear variables had a positive effect on the overall quality of the feature set. The retained variables, such as “Is

Table 2 Description of used attributes

Attribute	Description
Node ID	A unique identifier to distinguish the sensor node in any round and stage. Example: node number 25 in the third round and first stage → 001 003 025
Time	The current simulation time of the node
Is CH?	A flag indicating whether the node is a Cluster Head (CH) — 1 if CH, 0 if normal
Who CH?	The ID of the CH in the current round
RSSI	Received Signal Strength Indication between the node and its CH in the current round
Distance to CH	The distance between the node and its CH in the current round
Max distance to CH	The maximum distance between the CH and any node within the cluster
Average distance to CH	The average distance between nodes in the cluster and their CH
Current energy	The current energy level of the node in the current round
Energy consumption	The amount of energy consumed in the previous round
ADV CH send	Number of advertised CH broadcast messages sent to the nodes
ADV CH receives	Number of advertised CH messages received from CHs
Join REQ send	Number of join request messages sent by nodes to the CH
Join REQ receive	Number of join request messages received by the CH from the nodes
ADV SCH send	Number of advertised TDMA schedule broadcast messages sent to the nodes
ADV SCH receives	Number of TDMA schedule messages received from CHs
Rank	The position/order of this node within the TDMA schedule
Data sent	Number of data packets sent from a sensor node to its CH
Data received	Number of data packets received from the CH
Data sent to BS	Number of data packets sent to the Base Station (BS)
Distance CH to BS	The distance between the CH and the BS
Send code	The cluster sending code
Attack type	Type of the node, one of: Blackhole, Grayhole, Flooding, Scheduling, or Normal (if not an attacker)

Fig. 1 Correlation plot for the relation between the variables

CH”, “who CH”, and “Dist CH To BS”, maintained moderate VIF values, meaning they provide independent and valuable information for the model. This refined feature vector serves as a more robust input set, minimizing redundant relationships and improving both interpretability and predictive performance in subsequent analyses.

while CATBoost efficiently handles tabular and categorical data. The combination, optimized by metaheuristic algorithms, mitigates both bias and variance, common limitations of individual learners, resulting in robust and accurate intrusion detection [20, 21].

3 Methods

3.1 Voting meta ensemble

The proposed Voting Meta-Ensemble combines the predictive capabilities of multiple base learners to improve classification performance. Each independently trained model produces a probability distribution over the target classes. These outputs are then fused using soft voting, which averages the predicted class probabilities and selects the class with the highest average. This approach allows models with higher confidence to exert greater influence on the final decision, improving generalization compared to single models. In this framework, DNNs extract complex nonlinear features,

3.2 Deep neural networks

DNNs are employed for their ability to model complex nonlinear relationships in high-dimensional data. Each network consists of an input layer, several hidden layers, and an output layer. Hidden layers perform affine transformations followed by nonlinear activations, enabling hierarchical feature learning. A variable selection process ensures that only informative features are used, reducing overfitting and computational load. The DNN outputs class probabilities for intrusion detection, trained via gradient-based optimization and backpropagation [22, 23].

Table 3 Description of features and determination of selected features

NO	Feature	VIF score-initial feature vector	VIF score-reduced feature vector
1	Time	5.0682	4.8860
2	Is CH	3.6955	3.3497
3	who CH	3.5590	3.5374
4	Dist To CH	3.5336	2.3373
5	ADV S	1.2691	1.2538
6	ADV R	3.7796	3.1978
7	JOIN S	37.1710	Removed
8	JOIN R	2.9168	2.8654
9	SCH S	1.9004	1.8978
10	SCH R	27.8604	Removed
11	Rank	1.9650	1.6762
12	DATA S	4.3942	2.7054
13	DATA R	2.8153	2.8054
14	Data Sent To BS	1.7960	1.7755
15	Dist CH To BS	3.2483	3.2063
16	Send code	3.5184	3.0072
17	Expanded Energy	1.8788	1.8684

3.3 CAT-Boost classification (CATC)

CATBoost is a gradient-boosted decision tree algorithm optimized for categorical and tabular data. It reduces overfitting through ordered boosting and Bayesian estimation of feature contributions. In this study, CATBoost outputs class probabilities that are used alongside DNN outputs in the soft-voting ensemble. Equations detailing the internal computation of prediction values have been simplified to focus on how features are weighted and combined, rather than step-by-step algebra, improving clarity while retaining technical accuracy [24].

3.4 Metaheuristic optimization

Two population-based optimizers, Quadratic Interpolation Optimization (QIO) [25] and the Osprey Optimization Algorithm (OOA) [26, 27], are applied to fine-tune hyperparameters and decision thresholds.

- *QIO* emphasizes exploitation, allowing rapid convergence by approximating optimal parameter values using quadratic interpolation. Its fast convergence and local search efficiency make it suitable for refining decision boundaries in high-dimensional feature spaces.

- *OOA* focuses on exploration, mimicking the hunting behavior of ospreys to escape local optima and identify globally optimal parameter configurations.

Compared to classical optimizers such as Particle Swarm Optimization (PSO) or Genetic Algorithms (GA), QIO and OOA provide a complementary balance of convergence speed and exploration diversity. PSO often converges prematurely, while GA requires extensive iterations and can be computationally costly. By integrating QIO and OOA, the ensemble achieves faster, more reliable optimization with lower computational overhead.

3.5 Soft-voting ensemble operation

- DNN and CATBoost are independently trained on the feature set.
- Each model outputs a probability distribution over classes.
- Soft voting averages these probabilities, assigning the class with the highest mean probability.
- Optimized hyperparameters from QIO and OOA improve both model accuracy and ensemble reliability.

This design allows the ensemble to leverage the nonlinear representation power of DNNs, the categorical handling strength of CATBoost, and the efficient hyperparameter optimization of metaheuristics, yielding a robust intrusion detection system for WSNs.

3.6 Performance evaluation metrics

To evaluate how well the classification models work, the authors use several performance metrics that are common in the literature. These metrics measure different aspects of prediction quality, allowing for an extensive assessment of the models.

Accuracy represents the total fraction of instances that were correctly classified and can be expressed as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

True Positives (TP) mean positive cases that were identified correctly, and True Negatives (TN) denote negative cases that were correctly identified. False Positives (FP) are negative cases that were mislabeled as positive, and False Negatives (FN) are positive cases that were incorrectly labeled as negative.

Precision is the measure of the ratio of correctly identified positive cases to all predicted positive cases, which is a calculated value that shows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

Recall or True Positive Rate (TPR), as well as Sensitivity, defines the fraction of those positive cases that were identified correctly:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

F1 Score is one of the measures that combines the features of Precision and Recall, offering a balanced measure that reflects the errors that are both false positives and false negatives:

$$\text{F1 - score} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (4)$$

When classification performance is evaluated using various metrics, it provides evaluators with a comprehensive evaluation that considers true correctness, positive detection, and prediction reliability, particularly for imbalanced datasets where a single metric can lead to incorrect interpretation.

3.7 Model training and reproducibility

To ensure reproducibility and transparency, all models were trained with explicit hyperparameter settings, data split strategy, and preprocessing steps. The dataset was split into 80% training and 20% testing, with stratification applied to maintain the class distribution across splits. Table 4 shows the hyperparameters and architectural details for reproducibility.

The DNN architecture consists of a single hidden layer with 128 neurons, using ReLU activation for hidden layers and Sigmoid for output. Training was conducted with a learning rate of 0.1, a batch size of 32, and 100 epochs. The CATBoost classifier was configured with a depth of 6, learning rate of 0.15, L2 regularization of 3, and the RMSE loss function. The metaheuristic optimizers, QIO and OOA, were set with a population size of 30, search bounds defined per hyperparameter, and an iteration limit of 200. The stopping criterion was either convergence of the objective function or reaching the maximum number of iterations.

4 Result and discussion

This section presents the results from the ensemble-based intrusion detection framework, the proposal, and compares its performance with that of baseline models. The presentation of results first focuses on the convergence behavior of the optimization-enhanced ensembles, and a subsequent stage is the evaluation of their predictive performance by utilizing several classification metrics. Besides the overall model accuracy, the class-wise detection counts are also taken into

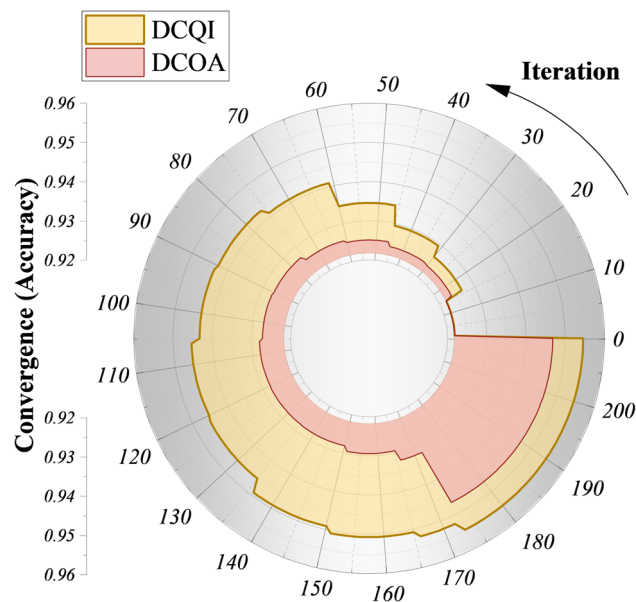
account, thus providing a deeper understanding of the interactions between precision, recall, and F1-score for different attack scenarios. The aspects of the model, such as stability, behavior, and sensitivity to features that result in performance patterns being linked to the latent optimization strategies, are revealed by convergence plots, heatmaps, ROC curves, and sensitivity analyses.

Figure 2 depicts a radial convergence plot that provides a visual comparison of the iterative performance trajectories of DCQI and DCOA, two hybrid ensemble configurations, over the entire process of optimization. The iteration progression and accuracy convergence are evident in the radial layout, which does not require detailed analysis, as each concentric arc represents incremental increases in classification accuracy. Right from the first iterations, DCQI (in yellow) goes to higher levels of accuracy faster, exceeding the value of 0.94 within the first few cycles of the optimization process, and also continuing with its upward trend until it reaches a level above 0.955, where it stabilizes. This means that the Quadratic Interpolation Optimization method is very effective in speeding up the convergence process by cutting off the decision boundaries in a more targeted and exploitation-driven way. Conversely, DCOA (in light red) traces a different pattern with a slower than DCQI convergence curve, where it is only after a long number of iterations that it becomes competitive at accuracy levels. Its final performance is marginally lower than that of DCQI, but the less steep slope signifies that OOA is more likely to employ the explorative search strategy, which can be one of the reasons why it is less prone to premature convergence in more complex optimization landscapes. The decrease of the performance gains in both models for the last iterations is very gradual, which is a strong indication of the saturation phase of the metaheuristic search, where the near-optimal solutions have been reached and the marginal improvements become smaller. The image as a whole highlights DCQI's advantages, including fast convergence and final accuracy, while also showing DCOA as strong in the exploration phase.

The computational efficiency of the proposed framework was carefully evaluated considering the resource constraints typical of WSNs. Experiments were conducted on a system with an Intel 11th Gen Core i7-1165G7 CPU @ 2.80 GHz and 16 GB RAM, using Python with a batch size of 512 for the DNN. The baseline DNN + CATBoost (DNCA) ensemble required approximately 35–36 min for training across 200 iterations, with soft-voting overhead being negligible. Its storage footprint was estimated at 93 MB. When metaheuristic optimization was incorporated, the training time increased due to multiple hyperparameter evaluations: the QIO-enhanced ensemble (DCQI) completed training in around 175 min (~ 2 h 55 min) with an estimated storage size of ~ 125 MB, benefiting from faster convergence via its exploitation-oriented search, whereas the OOA-boosted

Table 4 Hyperparameters and architectural details for reproducibility

Model	Hyperparameter					
	Layers	Activation functions	Learning rate	Batch size	Epochs	Random seed
DNN	128	ReLU, Sigmoid	0.1	32	100	
CatBoost Parameters	<i>Depth</i>	<i>Learning rate</i>	<i>L2 regularization</i>	<i>Loss function</i>		42
	6	0.15	3	RMSE		
Optimization Algorithms (QIO/OOA)	<i>Population size</i>	<i>Search bounds</i>				
	30	upper				

**Fig. 2** Radial plot illustrating the optimization process's convergence over iterations, showing continuous improvement in model performance

ensemble (DCOA) took approximately 190–195 min (~ 3 h 10–15 min) with 93 MB storage due to its exploration-driven search strategy.

To ensure the effectiveness and reliability of the intrusion detection system, the dataset was split into 80% training and 20% testing, while maintaining the original class distribution using stratification. Table 5 presents the number of samples per class for both training and testing phases.

Table 6 visually summarizes the quantitative evaluation of the base ensemble design (DNN + CAT; DNCA) and its metaheuristically improved versions, DNCA + QIO (DCQI) and DNCA + OOA (DCOA), by means of five canonical performance indices: accuracy, precision, recall, F1-score, and specificity, measured in both training and testing regimes. The DNCA model defines the reference framework, with an accuracy of 0.9218 (train) and 0.9217 (test), combined with high precision (0.9580/0.9557) and specificity (0.9807/0.9804), which are indicative of a conservative decision boundary that suppresses false positive

rates effectively. However, the recall values (0.9218/0.9217) indicate the potential for enhancement of sensitivity to minority or subtle attack classes. The addition of QIO (DCQI) to DNCA was the most effective in raising the performance level, reaching maximum training accuracy (0.9545) and testing accuracy (0.9562), as well as a balanced and raised F1-score (0.9593/0.9607), thus providing evidence of the optimizer's ability to adapt the decision surfaces in high-dimensional feature manifolds correctly. The positive predictive value increases significantly when precision values are close to one (0.9700/0.9703), whereas specificity metrics (0.9884/0.9892) reveal a stable performance under noise. The DCOA variation also presented a significant improvement over the baseline, characterized by accuracies of 0.9468 (train) and 0.9464 (test), with corresponding high F1-scores (0.9530/0.9527). This implies the effectiveness of OOA in improving generalization without overfitting, as evidenced by the negligible divergence between train and test performance. It is worth noting that precision was always higher than recall in all experiments, which points to a deliberate preference for false alarm reduction being the operationally most important feature in intrusion detection in resource-limited wireless sensor networks. The achievement of high accuracy, balanced F1-scores, and superior specificity in the metaheuristically augmented ensembles constitutes a strong argument in favor of exploiting the synergy between deep neural architectures, gradient-boosted decision trees, and sophisticated population-based optimizers to generate WSN security-context models that are stable, discriminative, and generalizable.

Figure 3 presents detailed views of the confusion matrix heatmaps for the baseline ensemble (DNCA) and its metaheuristically optimized variants (DCQI and DCOA) by showing the classification performance for all attack categories as well as for normal traffic. The main features of the diagonal are correctly classified instances, which, in the "Normal" category for all models, are very high. The highest number of true negatives (324,756) is reported from DCQI, which indicates that the latter is the best benign traffic detector. In line with this, DCQI lowers the number of false

Table 5 Class-wise Sample Distribution in Training and Testing Phases for the WSN Intrusion Dataset

Phase	Classes (attack type)				
	Normal: 0	Flooding: 1	TDMA: 2	Gray hole: 3	Blackhole: 4
Train	273,129	2851	5013	11,307	7433
Test	66,937	462	1626	3290	2617

Table 6 Evaluation metrics were employed to measure the predictive accuracy and overall performance of the models

Phase	Category	Models	Metrics				
			Accuracy	Precision	Recall	F1_score	Specificity
Train	Ensemble-Hybrid	<i>DNN + CAT (DNCA)</i>	0.9218	0.9580	0.9218	0.9335	0.9807
		<i>DNCA + QIO (DCQI)</i>	0.9545	0.9700	0.9545	0.9593	0.9884
		<i>DNCA + OOA (DCOA)</i>	0.9468	0.9668	0.9468	0.9530	0.9866
Test	Ensemble-Hybrid	<i>DNN + CAT (DNCA)</i>	0.9217	0.9557	0.9217	0.9331	0.9804
		<i>DNCA + QIO (DCQI)</i>	0.9562	0.9703	0.9562	0.9607	0.9892
		<i>DNCA + OOA (DCOA)</i>	0.9464	0.9658	0.9464	0.9527	0.9868

classifications when compared to DNCA in all adversarial scenarios. For instance, it can be seen from the heatmaps that the greatest differences in the numbers of false classifications between DCQI and DNCA occur in the “Flooding” and “TDMA” attack categories. QIO can enable the refinement of the decision boundary, thus providing a better distribution of the points and consequently reducing the number of false classifications. DCOA is able to show very good results, especially regarding “Blackhole” and “Grayhole” intrusions; however, the improvement in accuracy is somewhat less than that of DCQI, which is consistent with the optimization strategy that is more focused on the exploration of the solution space.

Figure 4 visually compares the on-hand and forecasted results via a column plot for the baseline ensemble model (DNCA) and its metaheuristically optimized counterparts (DCQI and DCOA) over different attack categories and normal traffic. The similarity between the measured and predicted bars allows a quick visual check of the respective models’ potential to reflect the underlying class distributions. When looking at the “Normal” class, all models demonstrate that the measured and predicted values are very close, with DCQI being the closest one, thus the best precision and recall. On the other hand, the DNCA model is quite off in its predictions for the “Flooding” and “TDMA” scenarios; both false and true counts are significantly underestimated. The optimized models, in particular DCQI, extend the contributions

of these discrepancies to a great extent; hence, they are able to improve decision boundary refinement and balance detection. Moreover, DCOA also works towards closing the gap between the on-hand and predicted counts, albeit to a lesser extent than DCQI, which is parallel with its slightly lower overall accuracy. Thus, the figure is a visual confirmation of the quantitative results, where the metaheuristic optimization, particularly QIO, is shown to be the most effective in achieving the predictive alignment with the actual network behaviors in wireless sensor network intrusion detection.

Table 7 breaks down the performance of the models across different attack scenarios, among which are the baseline DNCA ensemble and its metaheuristically improved versions, DCQI and DCOA. In the case of “Normal”, there is almost perfect precision for all models (> 0.998) with high recall (> 0.92), showing their ability to detect benign traffic with a high level of confidence. The DCQI model had the best F1-score value (0.9764) among all the others. This figure is the best indication that the model had a good balance between sensitivity and specificity. While adversarial conditions cause substantial variances, these kinds of variances are less pronounced in clean scenarios. In the “Flooding” setting, which represents an attack with a high level of noise and a high packet rate, the DNCA model’s precision was notably low (0.2942), whereas its recall was quite high (0.9218). It is the model’s habitual conduct, however, that it over-detects

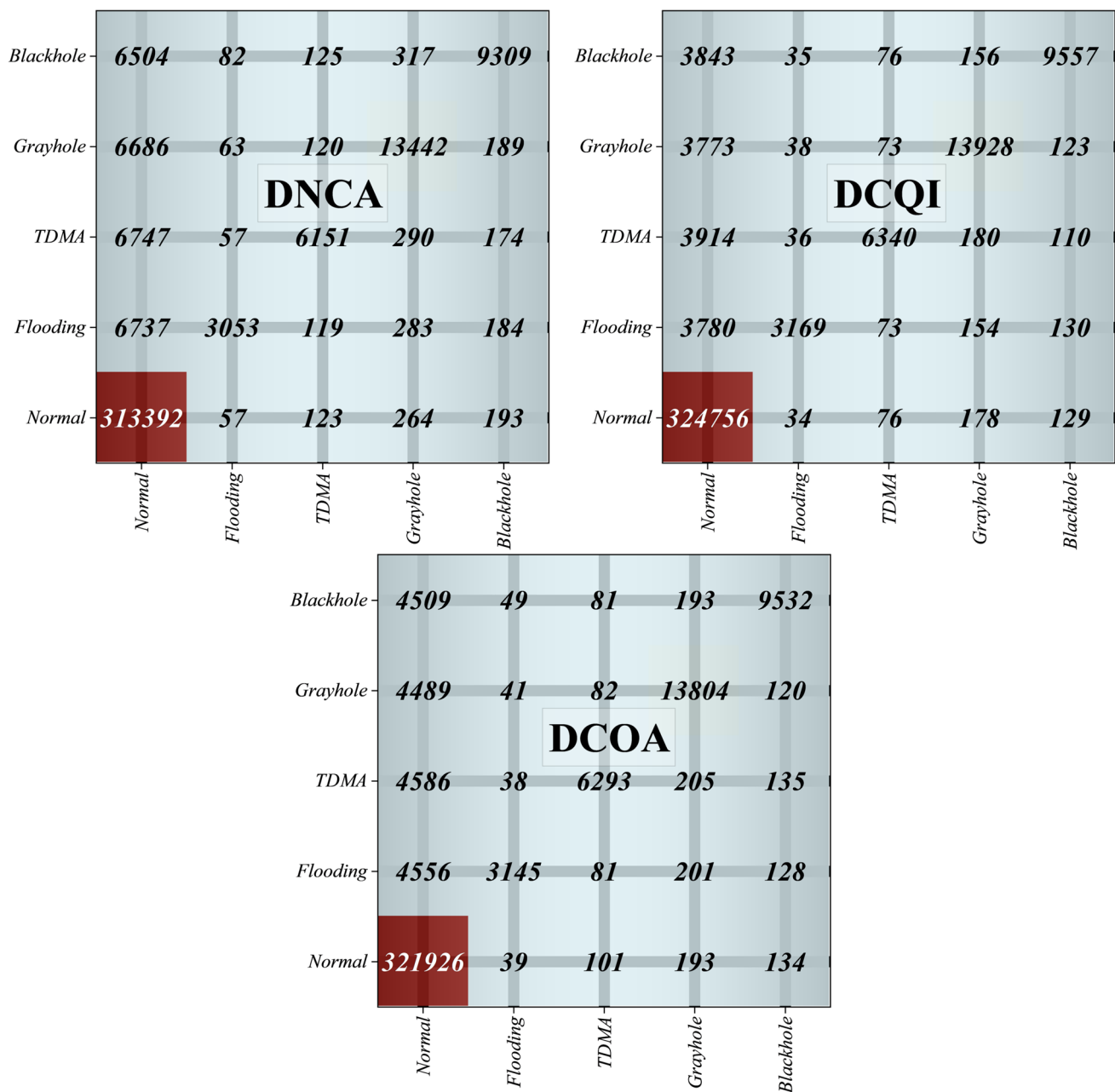
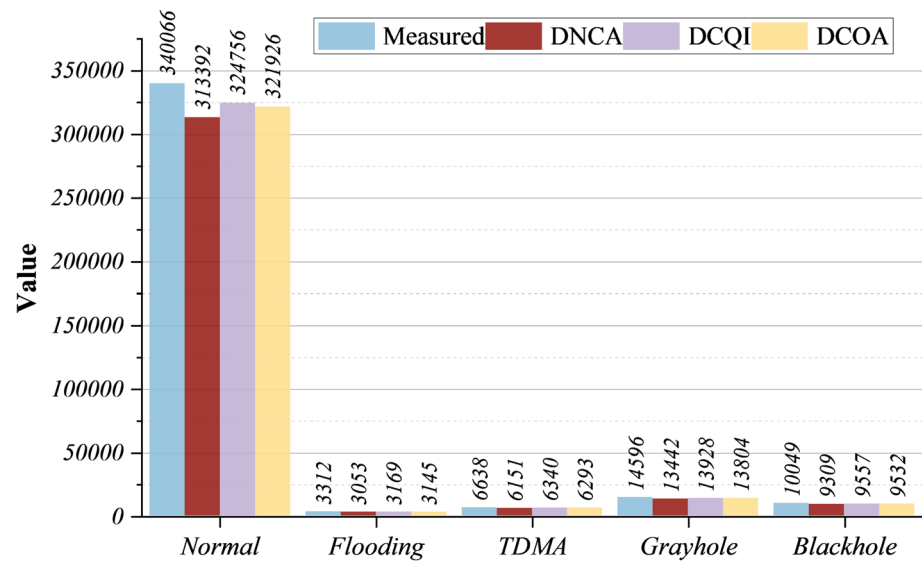


Fig. 3 Heatmap illustrating the classification performance of models across different classes

and thus generates a larger number of false positives. The precision of both DCQI and DCOA can be raised to a high level (0.4338 and 0.3877, respectively), with DCQI providing the highest F1-score (0.5969) as a result. Additionally, not only do the hybrid ensembles in the “TDMA” and “Grayhole” conditions contribute to a much higher precision, but also the F1-scores over DNCA, which is the key to the role that metaheuristic optimization plays in redefining the decision boundary. DCQI performance is also clearly observed in the “Blackhole” case, where packet-drop actions were the main features (precision: 0.6993, F1-score: 0.8060), while DCOA

can only achieve a limited improvement over the baseline. In general, DCQI is more likely than DCOA to attain higher precision and balanced classification metrics, suggesting that its optimization technique is more successful in locating the decision boundaries that are compatible with the different intrusion patterns in wireless sensor networks.

Figure 5 shows the Receiver Operating Characteristic (ROC) curves along with the corresponding Area Under the Curve (AUC) values for the baseline ensemble (DNCA) and the metaheuristically enhanced models (DCQI and DCOA).

Fig. 4 Column plot comparing measured and predicted values of models for different classes**Table 7** The models' performance was evaluated under various conditions

Category	Model	Condition	Metrics		
			Precision	Recall	F1 score
Ensemble	DNCA	Normal	0.9980	0.9216	0.9582
		Flooding	0.2942	0.9218	0.4461
		TDMA	0.4584	0.9266	0.6134
		Grayhole	0.6557	0.9209	0.7660
		Blackhole	0.5698	0.9264	0.7056
Ensemble-Hybrid	DCQI	Normal	0.9987	0.9550	0.9764
		Flooding	0.4338	0.9568	0.5969
		TDMA	0.5992	0.9551	0.7364
		Grayhole	0.7766	0.9542	0.8563
		Blackhole	0.6993	0.9510	0.8060
	DCOA	Normal	0.9986	0.9467	0.9719
		Flooding	0.3877	0.9496	0.5506
		TDMA	0.5590	0.9480	0.7033
		Grayhole	0.7447	0.9457	0.8333
		Blackhole	0.6636	0.9486	0.7809

The ROC curves are a complete measure of the models' differentiating capabilities by showing the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) for different decision thresholds. As the AUC values approach one, better classification performances are indicated along with a lower difficulty of differentiating between attack classes and benign traffic. The DCQI configuration is the one that achieves the highest AUC in all categories, thus demonstrating its capability of elevating the sensitivity levels without reducing the specificity, an immediate effect of the QIO-driven decision boundary optimization. Apart from that, DCOA also shows a significant increment compared to DNCA; however, the AUC values are somewhat lower

than those of DCQI, which agrees with the fact that its optimization strategy is more directed towards exploration. The baseline DNCA, which still has a high AUC in "Normal" and less complex attack types, decreases its performance in highly variable intrusion patterns such as "Flooding." In general, the diagram corroborates the numerical indicators by illustrating that metaheuristic integration, particularly QIO, achieves classifier robustness across various types of intrusion scenarios in wireless sensor networks.

Figure 6 presents a sensitivity analysis based on the cosine amplitude method (CAM) [28] that provides a quantitative visualization of the relative importance of each input feature to the predictive behavior of the suggested intrusion detection

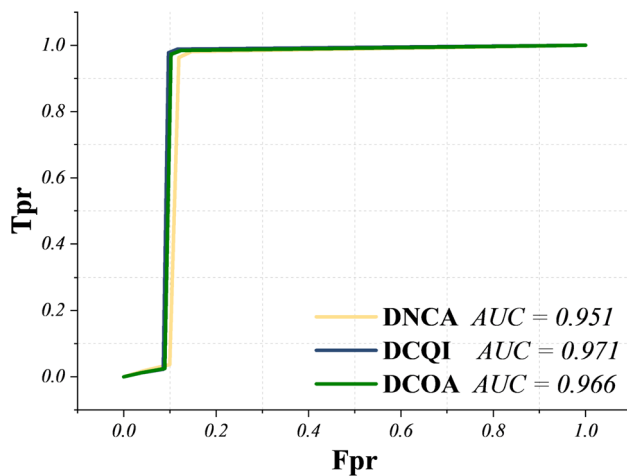


Fig. 5 ROC curves with corresponding AUC values illustrating the classification performance of the models

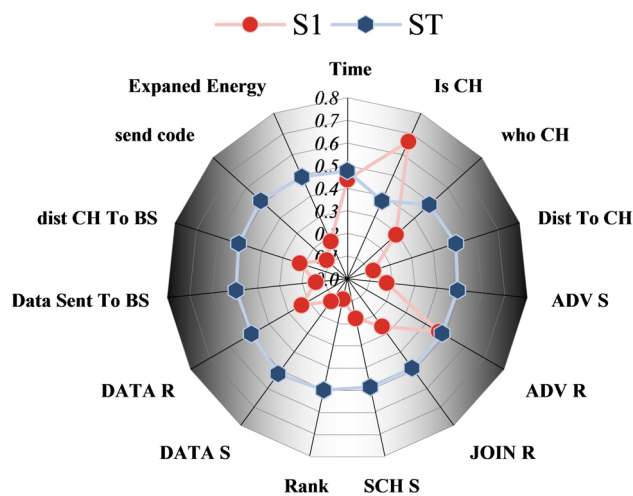


Fig. 6 CAM sensitivity analysis highlighting the contribution of each feature to the model's predictions

framework. Two complementary analytical methods have been applied to open up a clear and rigorous explanation of the feature contributions in the meta-ensemble introduced. The CAM characterizes the extent of agreement between each input feature vector and the output of the model's predicted class. To be more precise, for each feature, the cosine similarity is computed between its activation over the dataset and the predicted probability vector, thus offering a continuous relevance measure that indicates both the size and the direction of the feature's consistency with the model's decision. This technique enables the detection of features that are the main contributors to class discrimination, as well as handling the correlations existing in the feature space. The radar plot overlays two sensitivity profiles, S1 and ST, thus enabling direct comparison of feature contributions under different experimental setups. It is noteworthy that "Is CH"

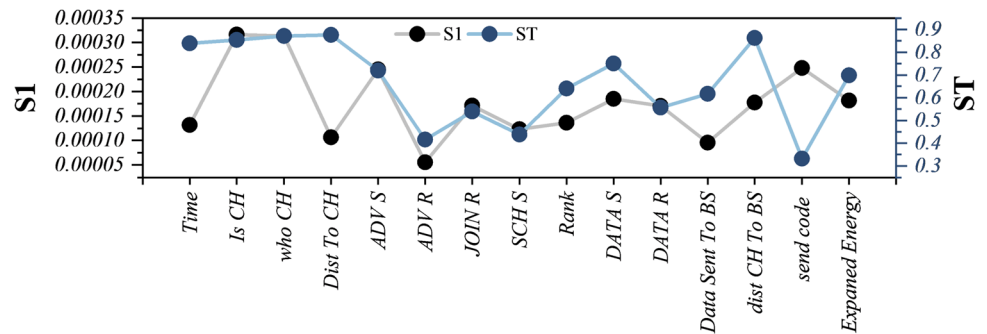
emerges as the most influential feature, with sensitivity values exceeding 0.75 in the S1 profile, indicating its decisive role in distinguishing cluster head-related behavioral patterns that are closest to attack detection. Besides, "Time," "Dist to CH," and "Who CH" are among the leading features that uncovered significant contributions in overcoming the limitations in topological and temporal dynamics in WSN communications.

On the other hand, the parameters such as "DATA R," "DATA S," and "Rank" have relatively low sensitivity values, based on which it can be suggested that these features have a few but minor contributions in the decision space. The departure between S1 and ST distributions implies that the model's sensitivity varies across different analytical settings, indicating an interaction between feature relevance and the adopted optimization or ensemble strategy.

Figure 7 represents the results of a quick sensitivity analysis, measuring the relative influence of each input feature on the model's output predictions. FAST sensitivity analysis [29] is a perturbation-based technique aimed at making efficient estimations of the impact of features on the output of the model. For each feature, its value is changed systematically within the observed range, and the model's predicted probabilities are checked for changes. After that, the sensitivity scores are combined and normalized to allow a ranking of feature importance to be drawn, which indicates those features that best affect the model's decisions. It uses a dual-axis depiction that shows S1 (black) on the left scale and ST (blue) on the right, giving a view of both absolute contribution and normalized importance at the same time. The research reveals that "Is CH," "Who CH," and "Dist to CH" are the three features with the highest sensitivity values in both S1 and ST, and hence the most influential in changing the properties of the classification boundaries in wireless sensor network intrusion detection. On the other hand, the "Data Sent to BS" and "Expanded Energy" attributes, showing substantial differences, pinpoint the consumption of energy together with the flow of data as the vital signs of the network's abnormal behavior. Moreover, the features "Rank" and "ADV S" have such a small influence on other features that their role in the network is expected to be that of a helper. The degree of similarity between S1 and ST is the largest intersection of feature trends in significance from different perspectives of the analysis, which constitutes their strongest point of agreement.

By comparing the two pictures, Figs. 6 and 7, the main characteristics of the two given methods of sensory analysis and their respective impact on the understanding of the feature importance are discussed. The CAM-based analysis (Fig. 6) gives a more spatially distributed picture of the relative importance of features such as 'Is CH', 'Who CH', 'Dist to CH', etc., in a multidimensional decision space, thus implying their structural and topological significance. Whereas, the fast sensitivity analysis (Fig. 7) is more detailed

Fig. 7 Fast sensitivity analysis showing the relative impact of each input feature on the model output



and absolute value-oriented, where S1 and ST impacts are shown, thus indicating the same key attributes' highest contributions as well as minor changes in the secondary features, "Expanded Energy" and "Data Sent to BS", for which the rest of the features are presumed to stay at a similar level. Besides, the CAM paradigm can be exploited for revealing the relative weightings in a holistic scenario, while the fast sensitivity technique makes it possible for the users to obtain the exact numerically-based rankings. As a result, these two methods together offer a richer, more corroborated, and more reliable account of feature dependencies in the model as well as feature stability across different analytical frameworks.

To ensure reliability and robustness, sanity checks were conducted. The analyses were repeated across multiple random seeds and on subsampled portions of the dataset, confirming that top-ranked features remained stable and low-impact features had negligible influence. Both CAM and FAST sensitivity consistently identified key structural and topological predictors, such as Is CH, Who CH, and Dist to CH, as the dominant drivers of model predictions. This dual-method approach not only validates the internal logic of the meta-ensemble but also enhances interpretability and trustworthiness, demonstrating that meaningful and explainable features underpin the model's high predictive performance.

5 Conclusion

This study has unveiled and solidly substantiated a novel intrusion detection framework for Wireless Sensor Networks (WSNs), whereby the integration of artificial deep neural networks, gradient-boosted decision trees, and metaheuristic optimization strategies has been implemented to unlock the latent capabilities of WSNs. Whereas the developed soft-voting meta-ensemble, consisting of Deep Neural Networks (DNNs) and CATBoost classifiers, had been thoroughly optimized by applying Quadratic Interpolation Optimization (QIO) and Osprey Optimization Algorithm (OOA), each supplying a different set of exploration and exploitation aspects to the indagation journey. The new arrangement is effectively

agnostic about the very hardware limitations, the nonstationarity of traffic, and the variability of the opponent's tactic entailed by the WSN intrusion problem in resource-limited environments. Extensive experimental assessments include a multi-metric evaluation protocol (accuracy, precision, recall, F1-score, specificity, and AUC) as evidence of the metaheuristically optimized ensembles outperforming the baseline model. The QIO-enhanced ensemble (DCQI) was notably the most performance-efficient across all metrics, characterized by fast convergence, high sensitivity to minority-class attack patterns, and great stability in the decision boundaries. The OOA-boosted (DCOA), while not outperforming DCQI in the last accuracy results, has proven to be more of an exploratory search behavior, thus lessening the danger of premature convergence and providing resilience in the more difficult optimization scenarios. Sensitivity of features to the model has been investigated by means of the Cosine Amplitude Method (CAM) and FAST feature ranking, not only to verify structural and topological predictors, notably Is CH, Who CH, and Dist to CH as the main driving factors, but also to suggest a stable pattern of feature importance across optimization strategies. This trust-enhancing factor opens up new opportunities for the administrator's intuitive problem-solving while also engaging in the explainable AI debate community in cybersecurity program/application fields. Notably, the proposed framework demonstrates robustness against unknown attacks and noisy data by leveraging probabilistic soft-voting, metaheuristic optimization, and feature sensitivity analysis. The ensemble can identify abnormal patterns even when attack signatures were not present in the training set, and its hierarchical feature representation allows resilience to random perturbations or sensor noise.

5.1 Limitations

Even though the results of this research show potential, the study still has some limitations that are worth mentioning. First of all, the dataset used, which is rich in features and includes multiple high-impact attack types, was created under the conditions of controlled experiments, so

the ability of the model to be generalized to large-scale, diverse, and fully dynamic WSN deployments is still empirically confirmed. Secondly, while metaheuristic optimization may be less time-consuming than other methods, it still requires offline training, which limits the possibility of real-time retraining in highly volatile environments. Thirdly, the range of attacks considered by the authors, although including major threats such as Blackhole, Grayhole, Flooding, and TDMA Scheduling, does not fully encompass the undetectable and evolving intrusion patterns (for instance, low-rate DoS, data exfiltration, and coordinated attacks across layers). Fourth, while the ensemble shows resilience to noisy or partially missing data, extreme levels of sensor failure or adversarial noise could still challenge detection performance. Lastly, this research is based on the assumption of the centralized detection model, while the distributed or federated detection systems, which can be more suitable for privacy-sensitive or geographically dispersed deployments, were not part of the study's focus.

5.2 Future directions

Due to these constraints, the researchers will concentrate on broadening the design to include online and adaptive learning paradigms. This will allow the system to change its decision boundaries on the fly as new and evolving attack vectors are encountered. Moreover, the adoption of federated learning might alleviate issues related to both scalability and privacy, while the use of lightweight deep learning (e.g., quantized or pruned networks) could make it easier for ultra-low-power sensor nodes to be deployed. Future work will also investigate enhanced anomaly detection strategies and noise-resilient learning to improve the detection of completely novel attacks and mitigate the impact of data corruption. Future works will focus on multimodal intrusion detecting, integrating cross-layer and contextual information that will improve the system's resistance to sophisticated and untraceable cyberattacks. To recap, this work offers a deep-ensemble-based intrusion detection system optimized with metaheuristics which is both computationally efficient and interpretable. Apart from being a potential avenue for the development of next-generation security systems for wireless sensor networks that are not only robust, resource-aware, and adaptable to the ever-increasing complexity and adversarial landscape of wireless sensing environments, this framework also signifies such a fund.

Authors contribution X.X. developed the algorithms, conducted experiments, and drafted the manuscript. W.D. supervised the study, refined the methodology, and edited the manuscript. All authors reviewed and approved the final version.

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Data availability Data are available from the author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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